Ex. 2.3 a)

Consider $\dot{x}(t) = -\lambda x(t)$, $x(0) = 1$, $t \in [0,1]$.

$$\frac{dx}{dt} = -\lambda x(t) \Rightarrow \int \frac{dx}{x} = \int -\lambda \, dt$$

$$\Rightarrow \ln|x| = -\lambda t + C$$

$$\Rightarrow x(t) = x_0 e^{-\lambda t} = e^{-\lambda t}.$$

We have $x_1 = \frac{1}{2}(1 + (1-\lambda h)^2) = 1 - \lambda h + \frac{\lambda^2 h^2}{2}$

$$x(h) = x(0) + \dot{x}(0)\cdot h + \frac{\ddot{x}(0)}{2!}h^2 + \frac{\dddot{x}}{3!}(\xi)\, h^3, \quad \xi \in (h_0, h_0+h)$$

$$x(0) = 1, \ \dot{x}(0) = -\lambda, \ \ddot{x}(t) = \frac{d}{dt} -\lambda x(t) = -\lambda^2 x(t), \ \ddot{x}(0) = -\lambda^2.$$

$$x(h) = 1 - \lambda h + -\frac{\lambda^2}{2!}h^2 + O(h^3)$$

now consider the discretization error

$$\cdot \ \tau(h) = \frac{x(h) - x_1}{h} = \frac{1 - \lambda h + -\frac{\lambda^2}{2!}h^2 + O(h^3) - 1 + \lambda h - \frac{\lambda^2 h^2}{2!}}{h}$$

$$= \frac{O(h^3)}{h} = O(h^2)$$

So Heun's method is a method of order 2.